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Roshan Kumar

Software Engineer

Portfolio: Portfolio
github.com/RoshanKumar404
LinkedIn:LinkedIn

Computer Science graduate with hands-on experience building low-level systems in Java, focusing on non-blocking network I/O and custom storage engines. Strong foundation in Data Structures, Concurrency, and Low-Level Design (LLD), actively seeking a Software Engineering role.

Skills

Languages	Java (Core, Concurrency, NIO), SQL, Python
Systems & Backend	Multithreading, Socket Programming, Java NIO (Selector, ByteBuffer), REST APIs, Reactor Pattern
Databases & Storage	PostgreSQL, PostGIS, LSM-Trees, Write-Ahead Logging (WAL), In-Memory Caching
Core CS	Data Structures, Algorithms, Operating Systems, Computer Networks, OOP (SOLID), Low-Level Design (LLD)
Tools & Platforms	Linux, Git, GitHub, Docker, JMH, wrk, JUnit, Postman

Projects

Asynchronous Reverse Proxy & Load Balancer | GitHub **2026**
Java, Java NIO, Concurrency, Socket Programming, Linux epoll

- Built a non-blocking Layer-7 reverse proxy in Java using **java.nio.channels.Selector** and the Reactor pattern to multiplex client connections over a fixed worker thread pool.
- Implemented bounded **ByteBuffer** pools and dynamic backpressure (toggling **OP_READ** on downstream sockets) to prevent memory spikes during slow upstream transfers.
- Integrated a lock-free **Token Bucket** algorithm to throttle incoming requests per IP, alongside Round-Robin and Least-Connections routing policies.
- Benchmarked using **wrk** against local HTTP upstreams, sustaining **10,000+ QPS** across 1,000 concurrent connections with sub-5ms average latency overhead.

Embedded LSM-Tree Key-Value Storage Engine | GitHub **2026**
Java, File I/O, Concurrency, Data Structures, Storage Internals

- Implemented an embedded key-value engine modeled after RocksDB, utilizing a **ConcurrentSkipListMap** MemTable and sequential disk flushes.
- Guaranteed write durability across process crashes by appending mutations to a **Write-Ahead Log (WAL)** using **FileChannel.force()** before memory commits.
- Built on-disk sorted **SSTables** with binary encoding, sparse index files, and custom **Bloom Filters** to eliminate disk reads for non-existent keys.
- Implemented background size-tiered compaction to merge sorted runs, discard tombstones, and verified crash recovery through JUnit crash-simulation tests.

Civic Incident Geospatial Backend | GitHub **2026**
Python, Flask, PostgreSQL, PostGIS, Geospatial Indexing

- Built a REST backend in Flask to ingest, validate, and query localized incident reports across dynamic radius boundaries.
- Designed spatial queries using **PostGIS** (**ST_DWithin**, **ST_MakePoint**) to aggregate nearby roadway incidents and eliminate duplicate user submissions.
- Added **GiST spatial indexes** on coordinate tables, cutting spatial lookup times on large test datasets down to under 100ms.
- Exposed aggregated endpoint data for frontend rendering using OpenStreetMap and density heatmap overlays.

Education

Bachelor of Technology in Computer Science and Engineering *Ramgarh Engineering College (Jharkhand University of Technology)*
2022 – 2026

Relevant Coursework: Data Structures, Algorithms, Operating Systems, DBMS, Computer Networks, Object-Oriented Programming, System Design

Higher Secondary (PCM)
Guru Gobind Singh Public School, Ranchi
2019 – 2021